

Congenital Anomalies of Kidney

Headings

1. • Renal Agenesis
2. • Duplication of Kidney
3. • Horseshoe Kidney
4. • Pancake Kidney
5. • Lobulated Kidney
6. • Anomalies of ascent
7. • PKD
8. • Accessory Renal Artery
9. • Wilms' tumour

1. Renal agenesis

- **Complete failure of development of kidney.**
- May be **unilateral or bilateral**.
- **Cause:** failure of formation of ureteric bud.
- GDNF signalling is important for ureteric bud growth; mutations involving **SALL1, PAX2, EYA1** may cause renal agenesis.

2. Duplication of kidney

- **Early division of ureteric bud** may produce an additional kidney.
- Kidneys may be **separate or fused**.
- Ureters may be **separate or partially fused**.

3. Horseshoe kidney ★

- The two kidneys are **fused at their lower poles** by an isthmus.
- The kidney remains at a **lower lumbar level**.
- **Reason:** ascent of the isthmus is prevented by the **inferior mesenteric artery**.

Memory:

Horseshoe → lower poles fused → low position → IMA blocks ascent.

4. Pancake kidney

- Both kidneys **fuse completely** to form a **single renal mass**.

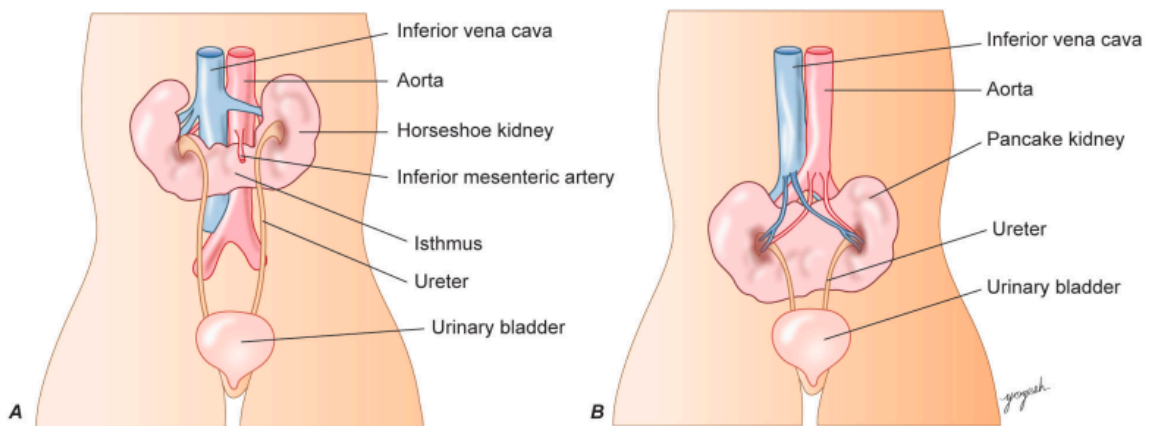


Fig. 20.10: Horseshoe kidney (A) and pancake kidney (B). Horseshoe kidney lies at the level of lower lumbar vertebrae as inferior mesenteric artery blocks ascent of isthmus of horseshoe kidney

5. Lobulated kidney

- Fetal kidney is normally **lobulated**.
- Normally, lobulation disappears during the **first year of life**.
- Persistence → **persistent lobulated kidney**.

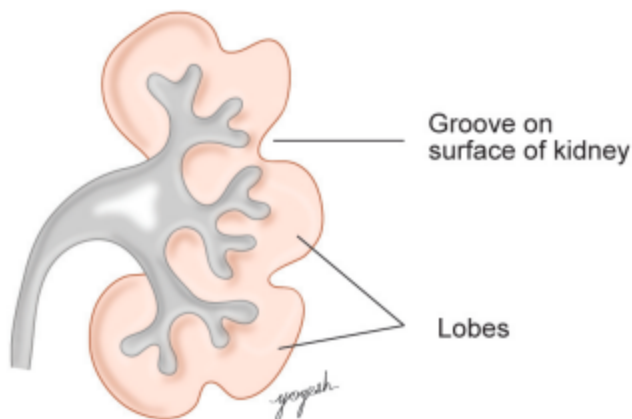


Fig. 20.11: Lobulated kidney. Usually, lobulation disappears in the first year of life. If it persists, it forms lobulated kidney

6. Anomalies of ascent

Anomaly	Position
Pelvic kidney	Remains in pelvic cavity
Lower lumbar kidney	Lies against lower lumbar vertebrae
Thoracic kidney	Excessive ascent into thoracic cavity

7. Congenital PKD (polycystic kidney disease) ★

- Characterized by formation of **multiple renal cysts**.
- Usually **bilateral**.
- Due to abnormal development/dilatation of **uriniferous tubules**.

Types:

- **ARPKD** — childhood presentation.
- **ADPKD** — usually presents in adulthood.

Treatment: Dialysis and kidney transplantation.

8. Accessory renal artery

- **Persistence of fetal renal arteries** produces accessory/supernumerary renal arteries.
- **Most common renal vascular anomaly**.
- May compress the ureter → **hydronephrosis**.

★ Very easy MCQ:

Most common renal vascular anomaly = Accessory renal artery

9. Wilms' tumour

- **Most common primary renal tumour of childhood**.
- Due to mutation of **WT1 gene**.

- Gene located on **chromosome 11p13**.



Last-minute recall

A D H P L A P A W

- **A** — Agenesis
- **D** — Duplication
- **H** — Horseshoe
- **P** — Pancake
- **L** — Lobulated
- **A** — Anomalies of ascent
- **P** — Polycystic kidney
- **A** — Accessory renal artery
- **W** — Wilms' tumour